

CONSTITUTION_ADDENDUM_QUALITY

Constitution Addendum: Quality Stack (Principle I Exception)

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Principle I — Container-First

| Every service runs in a container. No exceptions.

The quality stack (`docker-compose.quality.yml`) is an **opt-in additive compose file**, not part of the mandatory two-container product topology. It is governed by Principle I but does not alter the product's runtime contract.

Justification

Why This Is an Exception

The main `docker-compose.yml` defines exactly two containers (`qbittorrent`, `qbittorrent-proxy`) with `network_mode: host`. Adding SonarQube, PostgreSQL, and scanner containers to that file would:

1. Violate the product's minimal runtime topology
2. Require resources (2 GB+ RAM for SonarQube) that development machines may not have
3. Conflate CI/quality tooling with production services

Why It Complies with Principle I

All quality services **are** containerized. The additive compose pattern (-f docker-compose.yml -f docker-compose.quality.yml) preserves the base topology while allowing on-demand quality infrastructure. No quality service is installed on bare metal.

Trade-offs Accepted

Concern	Mitigation
SonarQube PostgreSQL credentials default to sonar/sonar	Quality stack never exposed externally; bound to 127.0.0.1
Scanner containers use host network	Matches project convention; scanners are run-to-completion, not long-lived
Additional compose file increases surface area	Profiles (run-once, quality) keep services off by default

Services Covered

- **sonarqube**: Code quality platform (community edition)
- **sonar-db**: PostgreSQL 16 for SonarQube persistence
- **snymk**: Dependency vulnerability scanner (requires SNYK_TOKEN)
- **semgrep**: Static analysis with SARIF output
- **trivy**: Filesystem vulnerability and misconfiguration scanner
- **gitleaks**: Secret detection in git history

Activation

```
$COMPOSE_CMD -f docker-compose.yml -f docker-compose.quality.yml  
up -d sonarqube  
./scripts/scan.sh --all
```

Decision Record

- **Date**: 2026-04-22
- **Decision**: Additive compose file for quality tooling, activated via profiles

- **Alternatives considered:** (1) Separate repo — rejected, splits codebase; (2) CI-only — rejected, developers need local access; (3) Bare-metal tools — rejected, violates Principle I

Principles Compliance Check (Phase 10 Verification)

Principle	Status	Evidence
I — Container-First	Compliant	All quality services containerized in additive compose; product topology unchanged (2 containers)
II — Manual CI	Compliant	<code>./ci.sh</code> remains the canonical CI entry point; GitHub Actions workflows are supplementary
III — No secrets in repo	Compliant	<code>.env</code> gitignored; CredentialScrubber filter active; Gitleaks scanner configured
IV — WebUI credentials	Compliant	<code>admin/admin</code> hardcoded and unchanged
V — Freeleech-only testing	Compliant	IPTorrents tests use <code>[free]</code> tagged results only
VI — TDD	Compliant	1,118 unit tests passing; coverage gate at 49%

Last verified: 2026-04-22 (Phase 10 final verification)